THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** Quadrant I	Check the signs of the abscissa and ordinate.
2	**(d)** 1	The degree is the highest power of the variable.
3	**(c)** $\sqrt{49}$	A rational number can be written as p/q with $q \neq 0$.
4	**(c)** $x^2 + 4x + 4$	$(a+b)^2 = a^2 + 2ab + b^2$
5	**(b)** 90°	An angle in a semicircle is a right angle.
6	**(c)** 84	Area $= l \times b$.
7	**(c)** $\frac{1}{2}$	$P(E) = \frac{\text{favourable}}{\text{total}}$ with 6 equally likely outcomes.
8	**(d)** 11	Put $n = 5$: $2 \times 5 + 1 = 11$.
9	**(a)** a number is divisible by 10	The part after ‘if’ is the hypothesis.
10	**(a)** 42	Sum 210 divided by 5.
11	**(d)** 549	Add column by column from the right, carrying 1 when a column sum exceeds 9.
12	**(c)** 100°	Angle sum of a quadrilateral is 360°.
13	**(a)** 2	$-4x + 2y = 6$ becomes $-4x + 2y - 6 = 0$.
14	**(d)** 224 cm^3	$lbh = 4 \times 8 \times 7$.
15	**(c)** 6	The abscissa is the x-coordinate.
16	**(c)** 11	Substitute x and simplify.
17	**(d)** $\sqrt{2}$	The square root of a non-square natural number is irrational.
18	**(b)** $x^2 - 8x + 16$	$(a-b)^2 = a^2 - 2ab + b^2$
19	**(b)** 7	Radius $= \frac{d}{2}$.
20	**(a)** 46	Perimeter $= 2(l + b)$.
21	**(c)** $\frac{1}{2}$	Two equally likely outcomes.
22	**(a)** 3	Subtract any term from the next term.
23	**(c)** its ones digit is 0	The part after ‘then’ is the conclusion.
24	**(d)** 40	Equal-sized groups: average of the averages.
25	**(a)** 8	Divisors of 24: 1, 2, 3, 4, 6, 8, 12, 24.
26	**(a)** 118°	Opposite angles are equal; adjacent angles add up to 180°.
27	**(c)** Yes	$5(5) + (7)(3) = 46$, equal to 46.
28	**(d)** 125 cm^3	5^3.
29	**(c)** -6	The ordinate is the y-coordinate.
30	**(b)** 5	Solve $p(x) = 0$.
31	**(a)** 0.75	Divide the numerator by the denominator.
32	**(d)** $x^2 - 9$	$(a+b)(a-b) = a^2 - b^2$
33	**(a)** 5	The perpendicular from the centre bisects the chord.
34	**(d)** 36	Perimeter $= 4\times$ side.
35	**(a)** $\frac{1}{2}$	$P = \frac{\text{green balls}}{\text{all balls}}$.
36	**(a)** 22	Add the common difference 5 to the last term.
37	**(b)** false	Counterexample: 2. 2 is prime and even.
38	**(b)** 87	29% of $300 = \frac{29}{100} \times 300 = 87$.
39	**(c)** 2	$38 = 12 \times 3 + 2$.
40	**(b)** 44 cm	Opposite sides are equal.
41	**(c)** 5	$6y = 34 - (2)(2)$ gives $y = 5$.

42	**(a)** 96 cm^2	$6a^2$.
43	**(a)** 5	$\sqrt{3^2+4^2}=5$
44	**(b)** 5	Put $x=0$: $y=c$.
45	**(b)** $0.\overline{3}$	Divide the numerator by the denominator.
46	**(b)** 961	$31=(30+1)$, apply $(a+b)^2$
47	**(d)** 25°	The angle at the centre is double the angle at the circumference.
48	**(d)** 48	Area $=\frac{1}{2}\times$ base $\times$ height.
49	**(b)** $\frac{1}{2}$	Count the multiples of the number.
50	**(c)** 16	Multiply the last term by the common ratio 2.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(b)** If a figure is a square, then it is a rectangle.	Subject becomes the hypothesis.
2	**(d)** 95	Total length is the sum of the components.
3	**(a)** 2	Common divisors: 1, 2.
4	**(c)** rhombus	Definition of a rhombus.
5	**(b)** a straight line	Fact.
6	**(a)** 84 cm^2	$2h(l+b)$.
7	**(c)** $(4, 6)$	Average the abscissae and the ordinates.
8	**(c)** 3	In $y = mx + c$ the slope is m.
9	**(b)** terminating	Recall the basic facts.
10	**(c)** 2499	$(50-1)(50+1) = 2500 - 1$
11	**(a)** 40°	Central angle $= 2\times$ angle at the circumference.
12	**(b)** 120	Area $=$ base $\times$ height.
13	**(d)** 0	Impossible events have probability 0; sure events have probability 1.
14	**(b)** 2	Divide a term by the previous term.
15	**(b)** 3	List the factors of 9 in pairs; the count is 3 (odd, so 9 is a perfect square).
16	**(c)** 51	$\dfrac{24\times 46 + 15\times 59}{39} = 51.$
17	**(d)** a step-by-step procedure that solves a problem	Definition.
18	**(b)** 20 cm	$\sqrt{12^2 + 16^2} = 20.$
19	**(c)** 4	Compare with $y = mx + c$.
20	**(d)** one-third of	$\frac{1}{3}\pi r^2 h$ versus $\pi r^2 h$.
21	**(b)** ordinate 0	Recall the properties of the axes.
22	**(b)** straight line	Recall the definition.
23	**(a)** 42	Apply Brahmagupta's rules for signed numbers.
24	**(a)** $(x+5)(x-5)$	$a^2 - b^2 = (a+b)(a-b)$
25	**(b)** 115°	Opposite angles of a cyclic quadrilateral are supplementary.
26	**(b)** 44	$C = 2\pi r.$
27	**(d)** 6	Count all possible outcomes.
28	**(d)** 21	$\frac{n(n+1)}{2} = \frac{6\times 7}{2}.$
29	**(d)** If a number is a multiple of 4, then it is a multiple of 8.	Interchange the hypothesis and the conclusion.
30	**(d)** 30 mL	$\frac{12}{100} \times 250 = 30.$
31	**(d)** 30	Sum of even numbers up to 11.
32	**(c)** 4 cm	Midpoint theorem: $DE = \frac{1}{2}BC$.
33	**(a)** parallel (or the same line)	Equal slopes.
34	**(d)** 2 cm	$\sqrt[3]{8} = 2.$
35	**(d)** $(3, -4)$	Only the sign of the ordinate changes.
36	**(a)** $3n + 2$	$a_n = a_1 + (n-1)d.$
37	**(a)** 14	$14^2 = 196$
38	**(c)** 49	Add first, then square.
39	**(b)** diameter	Recall the circle theorems.
40	**(d)** 154	$A = \pi r^2.$

41	**(b)** $\frac{5}{7}$	$P(\text{not } E) = 1 - P(E)$.
42	**(a)** an AP	Constant difference 2.
43	**(a)** 729	Digit sum of 729 is 18, a multiple of 9.
44	**(c)** total of all values divided by the total number of values	Use totals, not the plain mean of the two averages.
45	**(a)** gcd(n, m − n) for $m \geq n$	Common divisors divide the difference.
46	**(a)** parallelogram	Definition.
47	**(c)** 4	$1(2) + (7)k = 30$ gives $k = 4$.
48	**(b)** 462 cm^3	$\pi r^2 h$ with $r = 7$, $h = 3$.
49	**(d)** 7	Same ordinate: distance $= \lvert x_2 - x_1 \rvert$.
50	**(d)** $(0, 4)$	At $x = 0$, $y = c$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{7}{9}$	A rational number can be written as p/q with $q \neq 0$.
2	**(a)** $x^2 + 5x + 6$	$(x + a)(x + b) = x^2 + (a + b)x + ab$
3	**(c)** 90°	An angle in a semicircle is a right angle.
4	**(b)** 7	$r = \frac{C}{2\pi}$.
5	**(a)** $\frac{23}{50}$	Relative frequency $= \frac{\text{times it occurred}}{\text{number of trials}}$.
6	**(b)** 36	The sequence is n^2; $6^2 = 36$.
7	**(a)** True	The converse holds.
8	**(c)** 72	$\frac{79 \times 2 + 64 \times 1 + 69 \times 2}{5} = 72$.
9	**(c)** 1	Carries out of the columns: [0, 1].
10	**(b)** $12\sqrt{2}$ cm	$\sqrt{12^2 + 12^2} = 12\sqrt{2}$.
11	**(d)** $\frac{14}{3}$	Put x=0: $y = \frac{14}{3}$.
12	**(c)** 5544 cm^2	$4\pi r^2$ with $r = 21$.
13	**(b)** Quadrant II	Check the signs of the abscissa and ordinate.
14	**(a)** 3	The degree is the highest power of the variable.
15	**(b)** $\sqrt{3}$	The square root of a non-square natural number is irrational.
16	**(d)** $x^2 + 6x + 9$	$(a + b)^2 = a^2 + 2ab + b^2$
17	**(d)** 9	Radius $= \frac{d}{2}$.
18	**(c)** 135	Area $= l \times b$.
19	**(d)** $\frac{1}{3}$	$P(E) = \frac{\text{favourable}}{\text{total}}$ with 6 equally likely outcomes.
20	**(d)** 14	Put $n = 4$: $3 \times 4 + 2 = 14$.
21	**(a)** two lines are parallel	The part after ‘if’ is the hypothesis.
22	**(c)** 33	Sum 198 divided by 6.
23	**(b)** 561	Add column by column from the right, carrying 1 when a column sum exceeds 9.
24	**(b)** 55°	Angle sum of a quadrilateral is 360°.
25	**(a)** 4	$4x - 5y = 1$ becomes $4x - 5y - 1 = 0$.
26	**(a)** 162 cm^3	$lbh = 3 \times 9 \times 6$.
27	**(c)** -7	The abscissa is the x-coordinate.
28	**(d)** 14	Substitute x and simplify.
29	**(b)** 0.875	Divide the numerator by the denominator.
30	**(c)** $x^2 - 10x + 25$	$(a - b)^2 = a^2 - 2ab + b^2$
31	**(b)** 8	The perpendicular from the centre bisects the chord.
32	**(d)** 62	Perimeter $= 2(l + b)$.
33	**(d)** $\frac{1}{2}$	Two equally likely outcomes.
34	**(d)** 5	Subtract any term from the next term.
35	**(d)** it is a rectangle	The part after ‘then’ is the conclusion.
36	**(b)** 43	Equal-sized groups: average of the averages.
37	**(b)** 6	Divisors of 12: 1, 2, 3, 4, 6, 12.
38	**(a)** 72°	Opposite angles are equal; adjacent angles add up to 180°.
39	**(b)** No	$6(7) + (5)(4) = 62$, not equal to 40.
40	**(b)** 343 cm^3	7^3.

41	**(c)** 7	The ordinate is the y-coordinate.
42	**(c)** 3	Solve $p(x) = 0$.
43	**(b)** $0.\overline{6}$	Divide the numerator by the denominator.
44	**(c)** $x^2 - 16$	$(a+b)(a-b) = a^2 - b^2$
45	**(b)** 32°	The angle at the centre is double the angle at the circumference.
46	**(a)** 52	Perimeter $= 4 \times$ side.
47	**(d)** $\frac{1}{3}$	$P = \frac{\text{green balls}}{\text{all balls}}$.
48	**(d)** 25	Add the common difference 4 to the last term.
49	**(a)** false	Counterexample: 9. $9 = 3 \times 3$.
50	**(b)** 32	32% of $100 = \frac{32}{100} \times 100 = 32$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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